

Unit 1: Ratios and Proportions

Topic 1: Advanced Ratio Reasoning — Challenge Level 3 

Name: _____

Date: _____

- Q1.** The ratio of $A : B$ is 2:3 and $B : C$ is 6:5. What is the ratio $A : B : C$?
- (A) 2:6:5
(B) 4:6:5
(C) 2:3:5
(D) 4:3:5
- Q2.** In a mixture of 80 liters, the ratio of milk to water is 3:2. If 10 liters of water are added, what is the new ratio?
- (A) 3:3
(B) 4:3
(C) 12:13
(D) 1:1
- Q3.** A rectangle's length and width are in a 5:3 ratio. If the perimeter is 64 cm, what is the length?
- (A) 20 cm
(B) 40 cm
(C) 24 cm
(D) 12 cm
- Q4.** Simplify the complex ratio $\frac{2}{3} : \frac{4}{5} : 1$.
- (A) 10:12:15
(B) 2:4:5
(C) 6:8:15
(D) 10:8:15
- Q5.** A map scale is 1:250,000. If two cities are 8 cm apart on the map, how many kilometers are they apart in reality?
- (A) 20 km
(B) 200 km
(C) 2 km
(D) 25 km
- Q6.** If $3x = 5y$, what is the ratio of x to y ?
- (A) 3:5
(B) 5:3
(C) 8:5
(D) 15:1
- Q7.** On a coordinate plane, which point is NOT equivalent to the ratio 5:2?
- (A) (10, 4)
(B) (2.5, 1)
(C) (15, 6)
(D) (4, 10)
- Q8. [FRQ]** Use a tape diagram to solve: The ratio of boys to girls in a club is 5:3. There are 24 more boys than girls. Find the total number of members.
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- Q9. [FRQ]** Two numbers are in the ratio 4:7. If their sum is 132, find the smaller number. Show your algebraic steps.
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- Q10. [FRQ]** A birdhouse ratio of wings to beaks is 2:1. If you observe 120 wings and beaks in total, how many birds are there? Prove your answer.
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Chef's Tips

- **Combining Ratios:** To link $A : B$ and $B : C$, make the "B" value the same in both ratios.
- **Geometric Ratios:** In perimeter problems, remember that $P = 2(L + W)$. Use the ratio parts accordingly.
- **Unit Conversion:** 1 km = 100,000 cm. Keep this in mind for map scale problems.